Inverse Laplace transform

Rational functions and partial fractions

We seek the inverse transform of a rational function

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0},$$

where the coefficients are real and $a_n, b_m \neq 0$. In the terminology of the notes, F is **proper** if $m > n$, and **improper** if $m \leq n$. For an improper fraction, first use **polynomial long division**.

We consider four denominator-root cases: distinct real, distinct complex, repeated real, and repeated complex. Throughout, $u(t)$ is the unit step, $i^2 = -1$, and k^* denotes the complex conjugate of k.

A. Distinct real roots

To find the coefficient of $1/(s-p)$ at a simple root p of $D(s)$, multiply by $s-p$ and evaluate at $s = p$:

$$k = \lim_{s \to p}(s-p)F(s).$$

Example 15.1 – Distinct real roots. Find the partial-fraction expansion and inverse transform of

$$F(s) = \frac{96(s+5)(s+12)}{s(s+8)(s+6)} = \frac{k_1}{s} + \frac{k_2}{s+8} + \frac{k_3}{s+6}.$$

Compute k_1, k_2, and k_3 using the corresponding limits.

$$\begin{aligned} k_1 &= \lim_{s \to 0} sF(s) = \frac{96(5)(12)}{(8)(6)} = 120, \\ k_2 &= \lim_{s \to -8}(s+8)F(s) = \frac{96(-3)(4)}{(-8)(-2)} = -72, \\ k_3 &= \lim_{s \to -6}(s+6)F(s) = \frac{96(-1)(6)}{(-6)(2)} = 48. \end{aligned}$$

Thus

$$F(s) = \frac{120}{s} - \frac{72}{s+8} + \frac{48}{s+6}.$$

Check the expansion at the convenient points $s = -5$ and $s = -12$.

$$s = -5: \quad 0 = -24 - 24 + 48, \qquad s = -12: \quad 0 = -10 + 18 - 8.$$

These spot checks help catch algebraic errors; they do not prove the identity.

B. Distinct complex roots

The coefficient method is the same as for distinct real roots; the calculations now involve complex numbers.

Example 15.2 – A real root and a complex conjugate pair. Find the inverse transform of

$$F(s) = \frac{100(s+3)}{(s+6)(s^2+6s+25)}.$$

Factor the quadratic and write the partial-fraction form.

$$s^2+6s+25 = \underline{\qquad (s+3-i4)(s+3+i4) \qquad},$$

$$F(s) = \frac{k_1}{s+6} + \frac{k_2}{s+3-i4} + \frac{k_3}{s+3+i4}.$$

Compute the coefficients. Use conjugacy to check your work.

$$\begin{aligned}
k_1 &= \lim_{s\to -6}(s+6)F(s) = \frac{100(-3)}{25} = -12,\\
k_2 &= \lim_{s\to -3+i4}(s+3-i4)F(s) = \frac{100(i4)}{(3+i4)(i8)} = 6-i8,\\
k_3 &= \lim_{s\to -3-i4}(s+3+i4)F(s) = \frac{100(-i4)}{(3-i4)(-i8)} = 6+i8 = k_2^*.
\end{aligned}$$

Complex roots and their coefficients occur in $\underline{\quad \text{complex conjugate pairs} \quad}$, so only half of the complex coefficients need to be calculated independently.

Check at $s=-3$.

$$0 = \frac{-12}{3} + \frac{6-i8}{-i4} + \frac{6+i8}{i4} = -4 + (2+i1.5) + (2-i1.5).$$

C. Repeated real roots

For a root of multiplicity r, include all denominator powers from 1 through r. Derivatives of the product with the highest power determine the remaining coefficients.

Example 15.3 – A triple real root. Find the partial-fraction expansion and inverse transform of

$$F(s) = \frac{100(s+25)}{s(s+5)^3} = \frac{k_1}{s} + \frac{k_2}{(s+5)^3} + \frac{k_3}{(s+5)^2} + \frac{k_4}{s+5}.$$

First compute the coefficient at the simple root $s = 0$.

$$k_1 = \lim_{s\to 0} sF(s) = \underline{\qquad \frac{100(25)}{125} = 20 \qquad}.$$

Remove the triple factor and compute the required derivatives.

Define

$$G(s) = (s+5)^3 F(s) = \frac{100(s+25)}{s}.$$

Observe that $G(s) = \frac{k_1(s+5)^3}{s} + k_2 + k_3(s+5) + k_4(s+5)^2$. At $s = -5$,

- $G(s)$ is k_2; the other terms contribute zero.
- the first derivative of $G(s)$ is k_3; the other terms contribute zero.
- the second derivative of $G(s)$ is $2!\,k_4$; the other terms contribute zero.

So, find $G(s)$ in a form that is easy to differentiate, then find G' and G''.

$$G(s) = 100 + \frac{2500}{s}, \qquad G'(s) = -\frac{2500}{s^2}, \qquad G''(s) = \frac{5000}{s^3}.$$

Evaluate at $s = -5$ to find the coefficients.

$$k_2 = G(-5) = \underline{\quad -400 \quad}, \quad k_3 = \frac{G'(-5)}{1!} = \underline{\quad -100 \quad}, \quad k_4 = \frac{G''(-5)}{2!} = \underline{\quad -20 \quad}.$$

Write the expansion and check it at $s = -25$.

$$F(s) = \frac{20}{s} - \frac{400}{(s+5)^3} - \frac{100}{(s+5)^2} - \frac{20}{s+5},$$

$$s = -25: \qquad 0 = -\frac{4}{5} + \frac{1}{20} - \frac{1}{4} + 1.$$

D. Repeated complex roots

The repeated-root method is unchanged, but the differentiation and evaluation involve complex numbers.

Example 15.4 – A repeated complex conjugate pair.

$$F(s) = \frac{768}{(s^2+6s+25)^2} = \frac{768}{(s+3-i4)^2(s+3+i4)^2}.$$

Use the conjugate-pair form

$$F(s) = \frac{k_1}{(s+3-i4)^2} + \frac{k_2}{s+3-i4} + \frac{k_1^*}{(s+3+i4)^2} + \frac{k_2^*}{s+3+i4}.$$

Calculate the two coefficients at $s = -3+i4$.

Let

$$H(s) = (s+3-i4)^2 F(s) = \frac{768}{(s+3+i4)^2}.$$

$$k_1 = \lim_{s\to -3+i4} H(s) = \frac{768}{(i8)^2} = -12,$$

$$k_2 = \lim_{s\to -3+i4} H'(s) = \left.\frac{-2(768)}{(s+3+i4)^3}\right|_{s=-3+i4} = \frac{-1536}{(i8)^3} = -i3 = 3e^{-i\pi/2}.$$

Obtain the other coefficients by conjugation.

$k_1^* =$ -12, $k_2^* =$ $i3 = 3e^{i\pi/2}$.

Write the completed partial-fraction expansion.

$$F(s) = \frac{-12}{(s+3-i4)^2} + \frac{3e^{-i\pi/2}}{s+3-i4} - \frac{12}{(s+3+i4)^2} + \frac{3e^{i\pi/2}}{s+3+i4}.$$

E. Improper rational functions

Use polynomial long division to write an improper rational function as a polynomial plus a proper rational function.

Example 15.5 – Polynomial division before partial fractions.

$$F(s) = \frac{s^4 + 13s^3 + 66s^2 + 200s + 300}{s^2 + 9s + 20}.$$

Perform the long division. Identify the quotient and remainder.

$$F(s) = s^2 + 4s + 10 + \frac{30s + 100}{s^2 + 9s + 20}.$$

Indeed,

$$(s^2 + 9s + 20)(s^2 + 4s + 10) + (30s + 100) = s^4 + 13s^3 + 66s^2 + 200s + 300.$$

Expand the proper fraction and assemble the result.

$$\frac{30s + 100}{s^2 + 9s + 20} = \frac{30s + 100}{(s + 4)(s + 5)} = \frac{-20}{s + 4} + \frac{50}{s + 5}.$$

Thus

$$F(s) = s^2 + 4s + 10 - \frac{20}{s + 4} + \frac{50}{s + 5}.$$